PME Project Management (Glossary)

🧱 Section 1: Foundations of Project Management

What Is Project Management?

Project management is the structured application of knowledge, tools, and techniques to deliver a defined outcome within constraints of time, cost, and scope. It's not just about managing tasks—it's about aligning people, processes, and purpose.

Role of the Project Manager

- Leads the team toward shared goals
- Manages stakeholder expectations
- Balances the triple constraint:
  - ► **Scope** – What's included
  - ► **Time** – When it's delivered
  - ► **Cost** – Budget and resources

Project Life Cycle

1. **Initiation** – Define the project, identify stakeholders
2. **Planning** – Develop roadmap, schedule, budget, risk plan
3. **Execution** – Deliver work, monitor progress
4. **Closing** – Finalize deliverables, capture lessons learned

Methodologies

- **Waterfall** – Linear, phase-based approach
- **Agile** – Iterative, adaptive, feedback-driven
- **Lean** – Eliminate waste, maximize value
- **Six Sigma** – Reduce defects, improve quality
- **Hybrid** – Combine elements to suit context

Organizational Structures

- **Functional** – Departments manage their own work
- **Matrix** – Shared authority between PM and functional leads
- **Project-Oriented** – Teams formed around projects

PMO (Project Management Office)

- Sets standards and best practices
- Provides tools, templates, and training
- Supports governance and strategic alignment

🚀 Section 2: Initiation

SMART Goals

- Specific
- Measurable
- Achievable
- Relevant
- Time-bound

OKRs (Objectives & Key Results)

Objective : What you want to achieve

Key Results : How you measure success

Stakeholder Analysis

Use the Power/Interest Grid to classify stakeholders:

- High Power / High Interest → Manage closely
- High Power / Low Interest → Keep satisfied
- Low Power / High Interest → Keep informed
- Low Power / Low Interest → Monitor

Project Charter

A formal document that defines:

- Project purpose and scope
- Deliverables
- Roles and responsibilities
- Budget and timeline
- Risks and constraints
- Approval authority

Financial Fundamentals

- **Cost-Benefit Analysis** – Compare expected benefits vs. costs
- **ROI (Return on Investment)** – ROI = (Gain – Cost) ÷ Cost

🧭 Section 3: Planning

Work Breakdown Structure (WBS)

- Break project into manageable components
- Organize by deliverables, phases, or functions
- Supports scheduling, budgeting, and accountability

Scheduling

- **Critical Path Method (CPM)** – Identify tasks that determine project duration
- **Dependencies**:
  - ► Finish-to-Start (FS)
  - ► Start-to-Start (SS)
  - ► Finish-to-Finish (FF)
  - ► Start-to-Finish (SF)

Budgeting

- Estimate costs by task or phase

- Include contingency reserves
- Track actual vs. planned costs

Procurement

- Define what to buy and from whom
- Select vendors
- Write contracts
- Monitor delivery and performance
- Ensure ethical conduct

Risk Management

- Identify risks
- Analyze probability and impact
- Prioritize and plan responses
- Monitor and control risks
- Use tools like Risk Register and Fishbone Diagram

Communication & Team Management

- **RACI Chart**:
    - Responsible
    - Accountable
    - Consulted
    - Informed
- Plan meetings, updates, and feedback loops
- Clarify roles and expectations

🛠️ Section 4: Execution

Tracking Progress

Status Reports – Use RAG (Red, Amber, Green) indicators

Gantt Charts – Visualize schedule and dependencies

Burndown Charts – Track work remaining in Agile projects

Quality Management

- Plan quality standards
- Assure compliance
- Control deliverables
- Conduct UAT (User Acceptance Testing)

Data-Informed Decisions

- Use metrics to guide choices
- Ensure data privacy and ethical use
- Avoid bias in interpretation

Managing Change & Risk

- Escalate issues appropriately
- Use ROAM Analysis:
  - ► Resolved
  - ► Owned
  - ► Accepted
  - ► Mitigated
- Provide “Air Cover” to protect the team from external pressure

🔄 Section 5: Agile & Scrum

Agile Manifesto

Values:

- Individuals and interactions > Processes and tools
- Working software > Comprehensive documentation
- Customer collaboration > Contract negotiation
- Responding to change > Following a plan

Principles:

- Deliver frequently
- Welcome change
- Build projects around motivated individuals
- Reflect and improve regularly

<u>Scrum Roles</u>

Scrum Master – Facilitates process

Product Owner – Owns backlog and vision

Developers – Build and deliver increment

<u>Scrum Artifacts</u>

Product Backlog – List of desired features

Sprint Backlog – Tasks for current sprint

Increment – Completed, usable product

<u>Scrum Events</u>

Sprint – Time-boxed development cycle

Sprint Planning – Define sprint goal and tasks

Daily Scrum – 15-minute team sync

Sprint Review – Demo and feedback

Retrospective – Reflect and improve

✅ Section 6: Closing

Project Closure

- Confirm deliverables
- Obtain stakeholder sign-off
- Release resources

Post-Mortem

- Document lessons learned
- Identify successes and failures
- Recommend improvements

Impact Reporting

- Use metrics to show value
- Visual storytelling to communicate results
- Include ROI, customer satisfaction, and performance indicators

🤖 Section 7: Career Development with AI

AI in Job Search

- Resume writing with X-Y-Z formula
- Practice interviews with AI tools
- Tailor applications using job description analysis

AI in Project Work

- Draft charters
- Identify risks
- Map stakeholders
- Summarize meetings and decisions

Ethical AI Use

- Maintain human oversight
- Protect data integrity
- Disclose AI involvement responsibly